RESEARCH BY ABRAHAM SU

# The Cognitive Lab

90% of college students now use AI. But what happens to the human "cognitive musculature" when we outsource our thinking?

Launch Probe

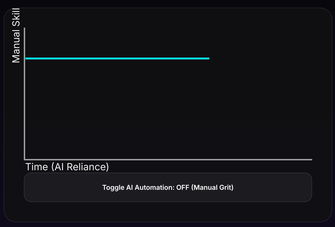
## The Pilot's Paradox

**Manual Skill vs. Automation**

The paper draws a stark parallel to modern pilots. When automation takes over, manual flying skills atrophy—leading to crisis when the system fails.

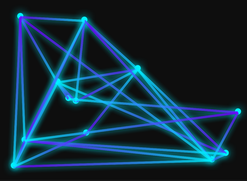
90%

LLM USAGE IN HIGHER ED




Toggle AI Automation: OFF (Manual Grit)

## Neuroplasticity vs. Outsourcing



**Learning Mode**

Toggle between deep effortful learning and rapid AI outsourcing to see how your neural pathways react.

Effortful Learning

AI Outsourcing

Neural Activity Index

100%

**Current State:** Effort strengthens synaptic connections through brain plasticity.

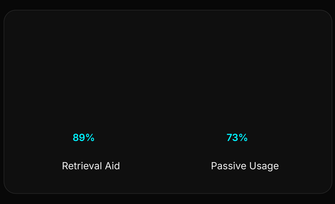
## The Retention Challenge

**The 89% vs 73% Reality**

Researchers An et al. (2025) found that **effortful retrieval** (testing yourself) beats **passive reading** with AI assistance.

Want to test your own brain plasticity?

Start Retrieval Test




## The Atrophy Simulator

**Cognitive Inputs**

Deep Reading Effort

Analytical Writing

AI Reliance (Outsourcing)

**Neural Plasticity Status**

Active

Balanced growth. You are maintaining your cognitive edge.

Based on "AI in Education: Where Does it Add Up?" by Abraham Su

Antigravity Premium Design Prototype